Computer Communication and Networks

Data Link Layer: Error: types, detection, correction, detection and correction technique

Session 8

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Data Link Layer

Learning Outcome:- At the end of the lesson, you are expected to be able to:

- Identify the types of Error.
- Evaluate the error detection and correction techniques.
- Demonstrate the different technique of error detections.



Data Link Layer: Introduction

Introduction:

- When data transfer from one device to another, **the data can be corrupted during transmission.**
- The data corruption can take place due to environmental interference and physical defects in the communication medium.
- For **reliable communication**, errors must be detected and corrected.
- Error detection and correction are implemented either at the **data link layer** or the **transport layer** of the OSI model.

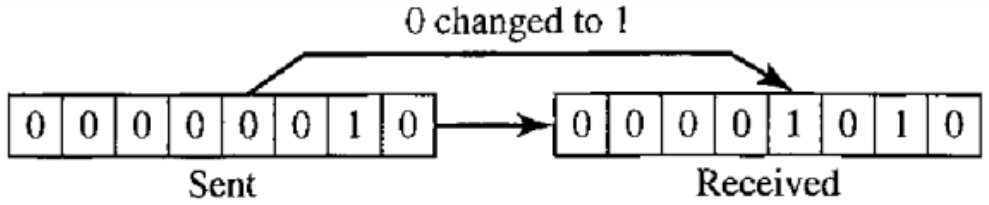
- The interferences can change the **timing** and **signal**. If the signal is carrying binary encoded data, such changes can altered the meaning of the data (error occurs).



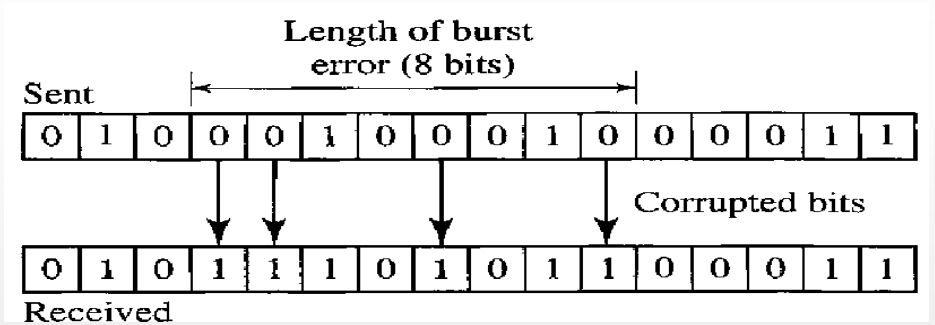
Data Link Layer: Types of error

❖ Types of Error:

➢ These errors can be divided into two types:



1. **Single-bit Error**
   - ✓ Only one bit in the data unit has changed (from 1 to 0 or from 0 to 1)
   - ✓ Mostly occurs in parallel transmission.



2. **Burst Error**
   - ✓ Two or more bits in the data unit have changed.
   - ✓ Most likely to happen in a serial transmission.
   - ✓ The length of burst is measured from the first corrupted bit to the last corrupted bit.



Data Link Layer: Error Detection

❖ Error Detection Code:

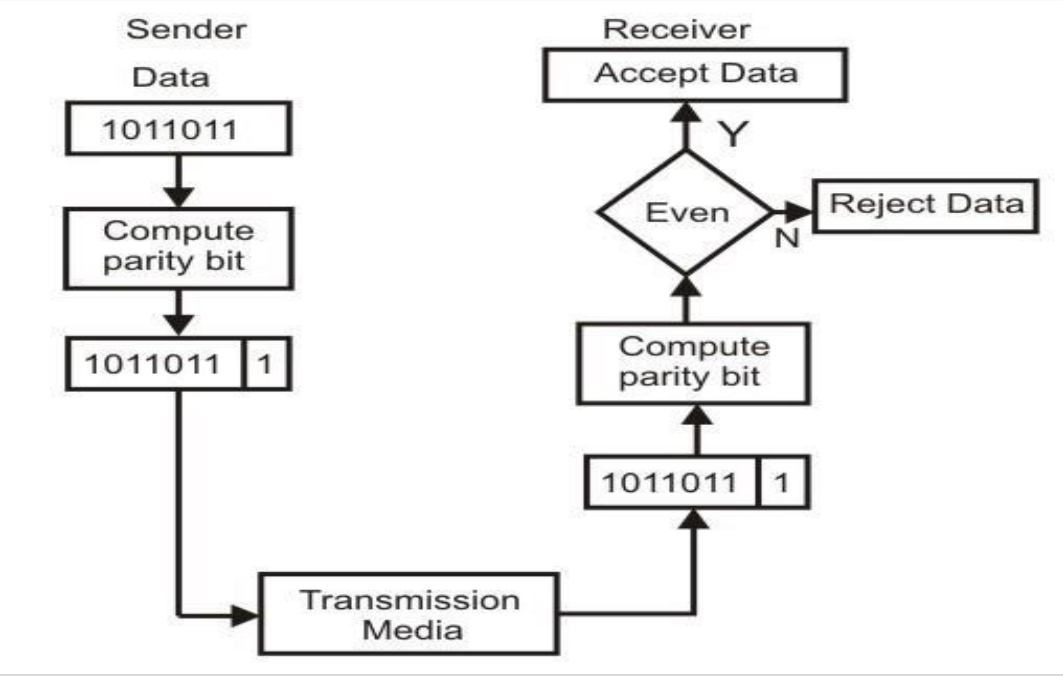
- **Basic approach** used for error detection is the use of redundancy.
- Redundancy is an **extra bits added to facilitate detection and correction of errors.**
- **Popular techniques are**:
  1. Simple Parity Check
  2. Two-dimensional Parity Check
  3. Cyclic Redundancy Check
  4. Checksum



Data Link Layer: Error Detection

❖ Simple Parity Check or One-Dimensional Parity Check

- Most common and least expensive mechanism of error detection .
- In this technique, **a redundant bit** called parity bit is appended to every data unit so that the total number of 1's in the unit (including the parity bit) becomes even.
- **Two types:**
  1. VRC (Vertical Redundancy Check)
  2. LRC (Longitudinal Redundancy Check)



Even Parity Checking Scheme

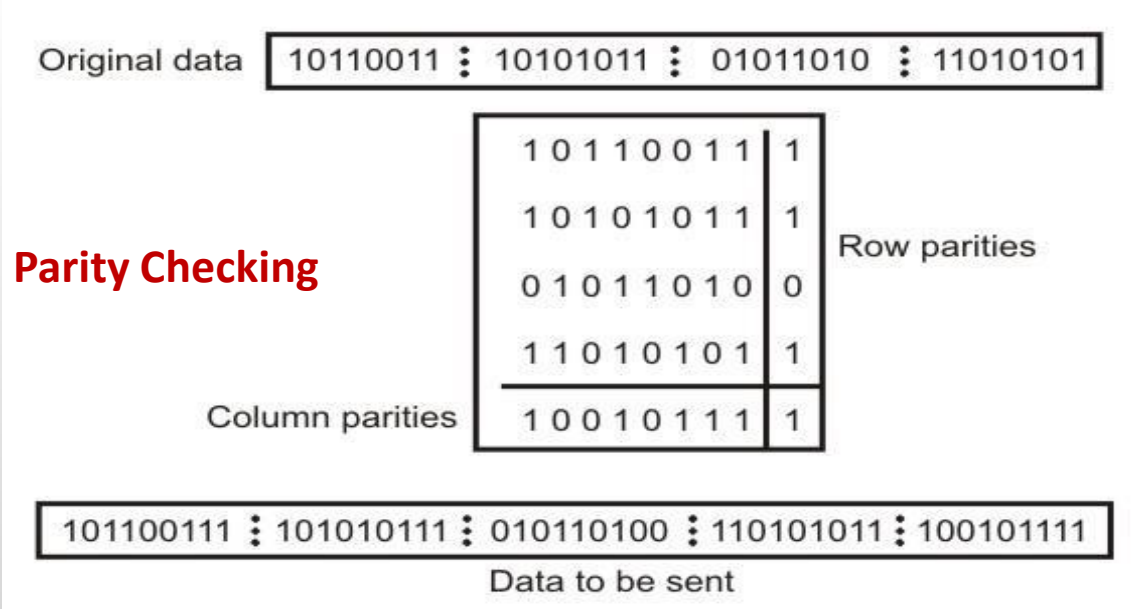


Data Link Layer: Error Detection

❖ Two-Dimensional Parity Check:

- Performance can be improved by using two-dimensional parity check, which organize the block of bits in the form of a table.
- Parity check is calculated in each ROW and all COLUMNS and both are send along with the data. At the receiving end these are compared with the parity bits calculated on the received data.

Two-dimensional Parity Checking





Error Detection

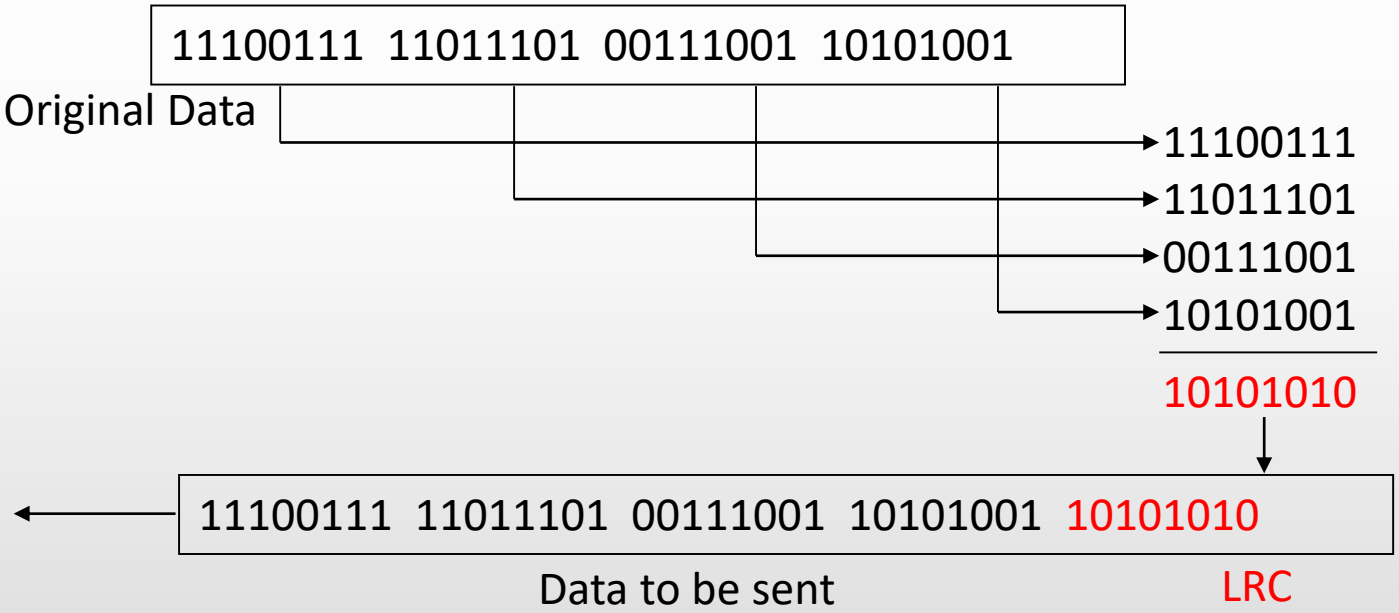
- **Error Detection**
  - Vertical Redundancy Check (**VRC**)
    - Append a single bit **(0 or 1)** at the end of data block such that the number of ones is even.
      - ✓ **Even Parity** (odd parity is similar)
        - 0110011 → 01100110
        - 0110001 → 01100011
  - **VRC** is also known as **Parity Check**
  - Performance:
    - Detects all odd-number errors in a data block



Error Detection

Error Detection

- Longitudinal Redundancy Check **(LRC)**
  - Organize data into a table and create a parity for each column

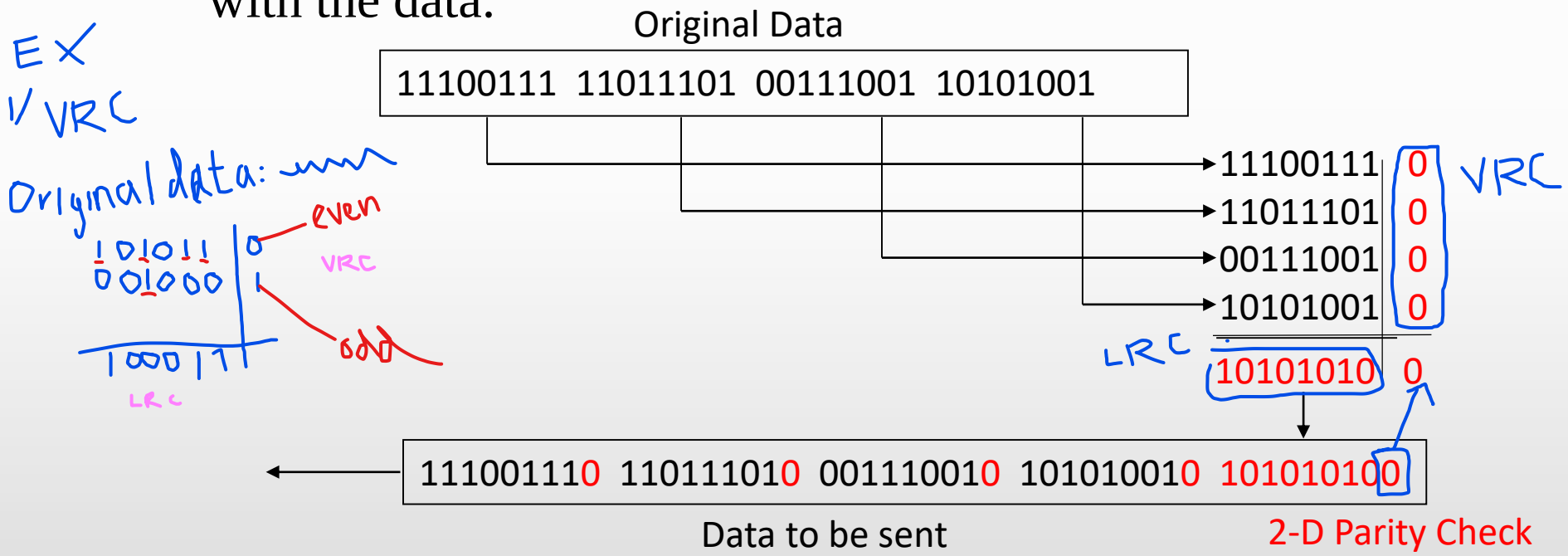




Error Detection

Error Detection

- 2-D Parity Check
  - Parity check is calculated in each row and all columns and both are send along with the data.





Error Detection

Error Detection

- You want to sent “OSI” string to the receiver through unreliable link. In order to detect an error, perform the following:
  1. **VRC**
  2. **LRC**
  3. **2-D Parity Check**

A-Z=65-90

a-z=97-122

Solution:

Steps:

1. Find the ASCII equivalent value of capital O
2. Find the ASCII equivalent value of capital S
3. Find the ASCII equivalent value of capital I
4. Convert each letter from decimal number into binary digit
5. Then detect the error accordingly.



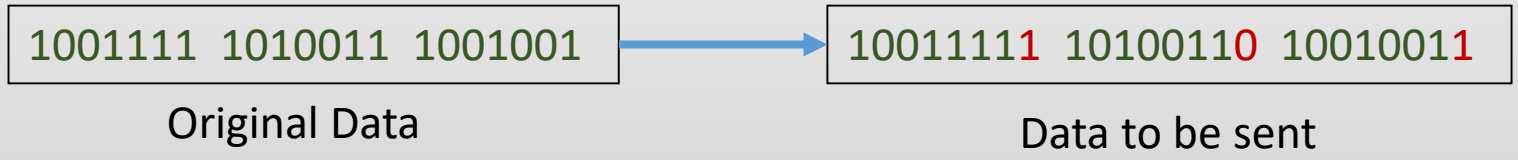
Error Detection

ASCII equivalent

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90

- O=79, S=83, I=73
- O=1001111
- S=1010011
- I=1001001

1. VRC

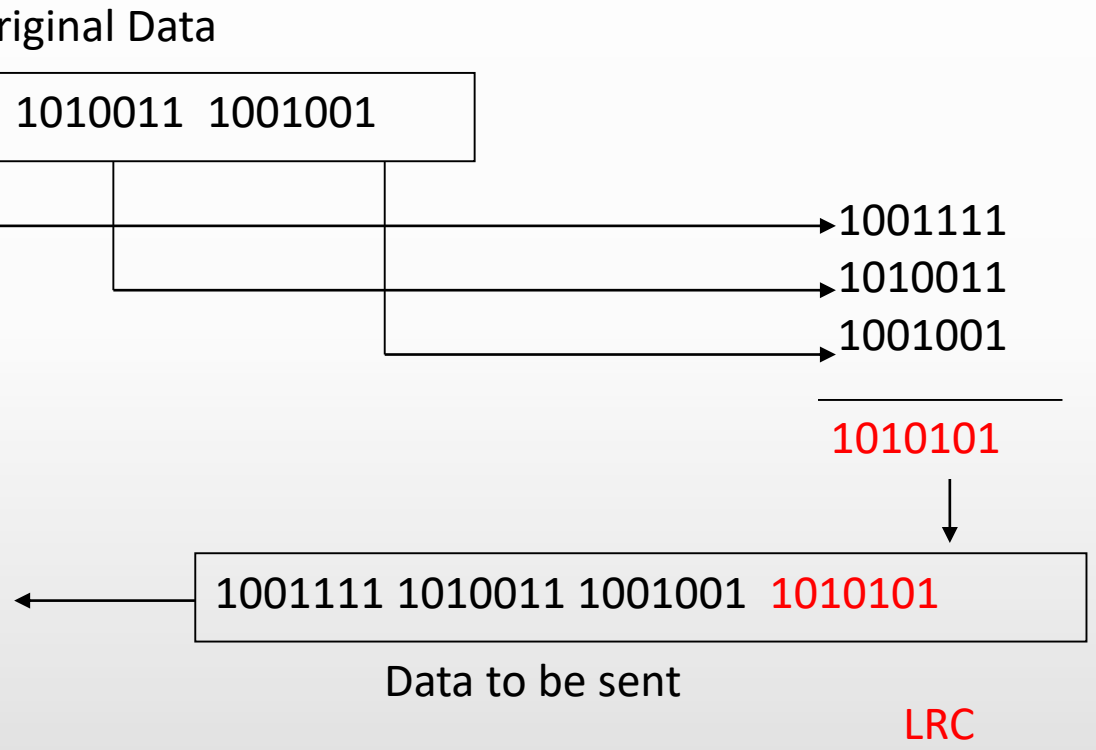




Error Detection

- **Error Detection**

  2. Longitudinal Redundancy Check (**LRC**)

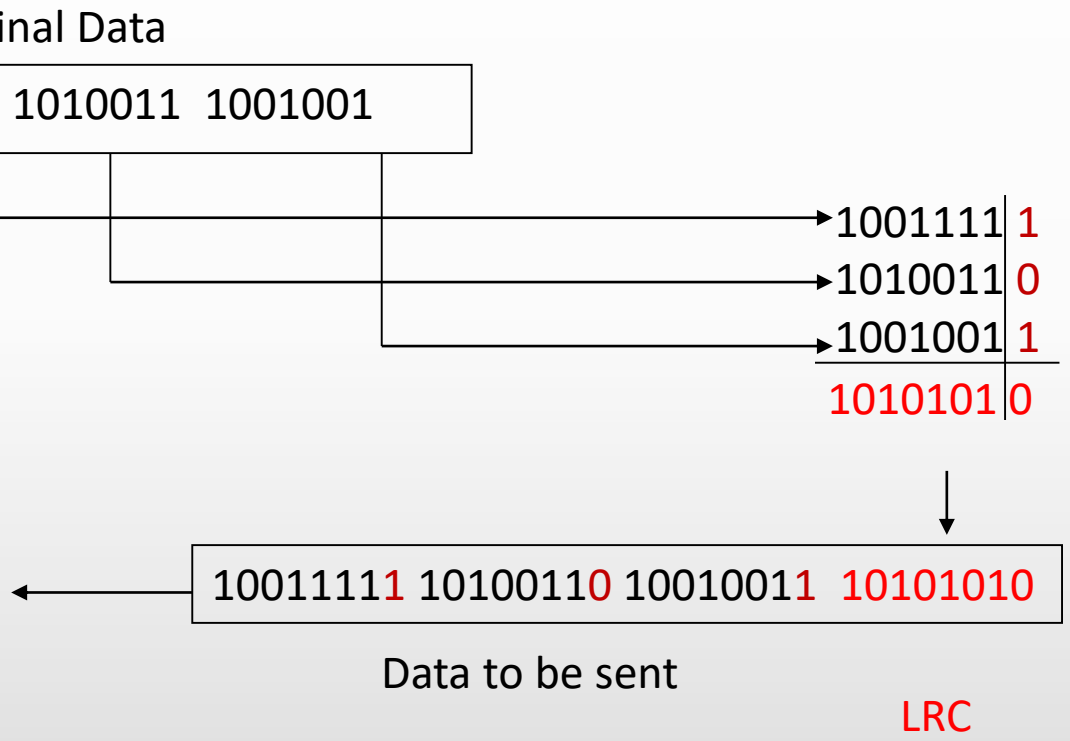




Error Detection

- **Error Detection**

  3. 2-D Parity Check





Error Detection

- The string “HeLlo” is sent through unreliable link. In order to detect an error, perform the following:
  1. **VRC**
  2. **LRC**
  3. **2-D Parity Check**

Solution: ??

H=72 →(1001000)

e=101 →(1100101)

L=76 →(1001100)

l=108 →(1101100)

o=111 →(1101111)



Error Detection

- VRC

H=1001000

e=1100101

L=1001100

l=1101100

o=1101111

1001000 1100101 1001100 1101100 1101111

10010000 11001010 10011001 11011000 11011110

1001000 1100101 1001100 1101100 1101111

Original Data

- **LRC**

1001000
1100101
1001100
1101100
1101111
1100010

Data to be sent

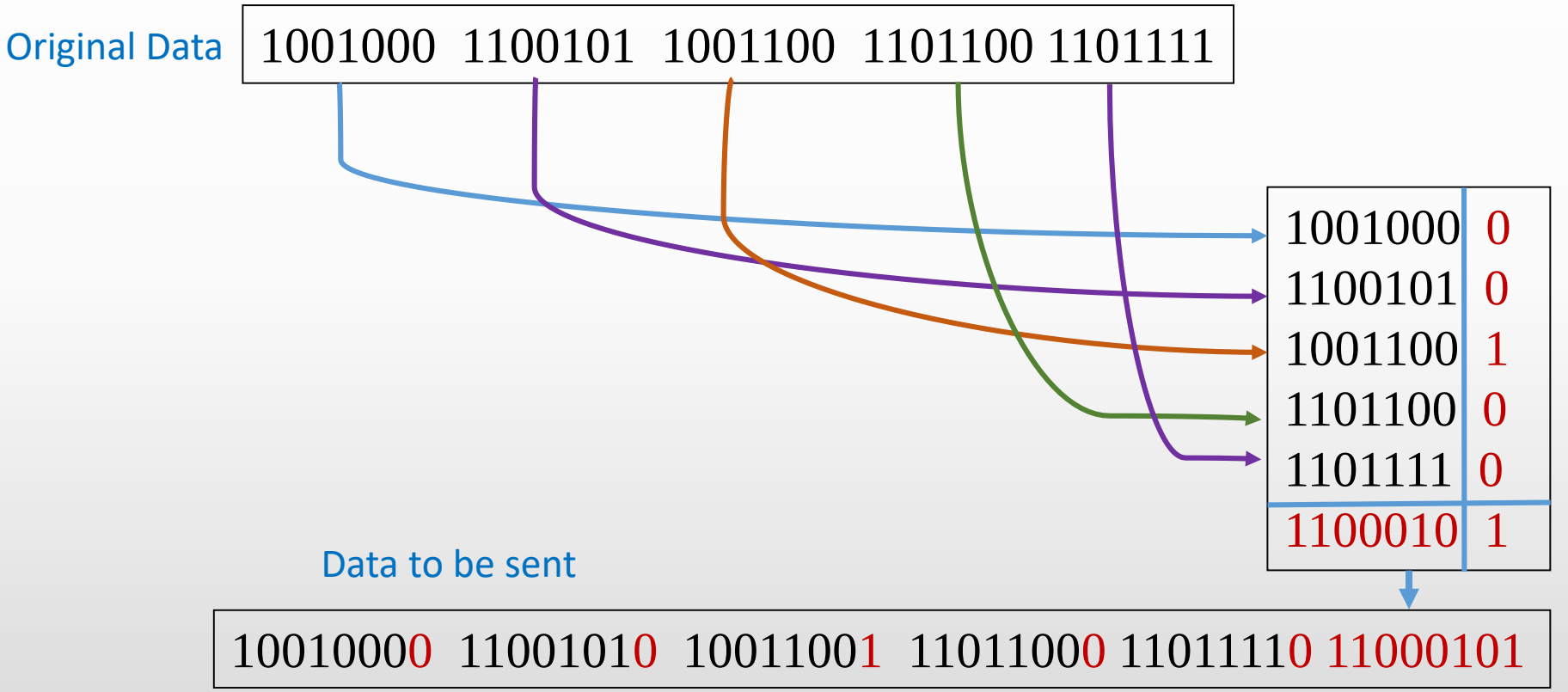
1001000 1100101 1001100 1101100 1101111 1100010

LRC



Error Detection

- 2-D Parity Check





References

References:

1. Stallings, W., Data & Computer Communication, Prentice-Hall
2. Tanenbaum, Computers Networks, PHI
3. Forouzan, B.A., Data Communication & Networkings, 4th Edition.
4. Mr. Yeshi Wangchuk's CCN Class Slides (2014), College of Science and Technology, Rinchending, Phuentsholing, Bhutan



See you in next class.